

Foundations of Artificial Intelligence - 2026.1

Environments

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Course Outline

1. Environments, Agents, and Rationality
2. Agents
3. Rationality

Environments

The heart of any AI system is the **environment**.

An environment is the world of interest with its rules of transition:

- The world is in exactly one out of many (possibly infinite) **states**.
- The environment evolves upon **events**.

The description of an environment are the *laws* that govern the world of interest.

Inside the environment itself there is no notion of time (unless encoded explicitly).

Environments

Examples

Examples of environments are:

- A chess board.
- A patient diagnosis or treatment situation.
- Vehicles in a street network.
- An economy.

Environments can be **virtual** or **physical** (or both); this is not relevant for the formalism.

Environments

State Transition

An **environment** $\langle S, E, \psi \rangle$ consists of

- a set S of states,
- a set E of events, and
- $\psi : S \times E \rightarrow S$ is a (possibly stochastic) **state transition function** that defines which state is reached given a previous state and an event.

An **environment instance** $\langle S, E, \psi, s_0 \rangle$ is an environment together with an **initial state** s_0 .

Environments

Formal Example

A very simple example is an environment where

- $S = E = \mathbb{N}$
- $\psi(s, e) = s + e$

maybe with $s_0 = 0$.

Several questions arise here:

- 1 *When* do the events occur?
- 2 *Who* decides which event occurs?
- 3 *Where* is the state information maintained and transitions executed?

Environments

The Environment Loop

The environment is a *structural* and not an *procedural* object: It does not execute anything nor is it associated with time.

To obtain a dynamic process, use an **environment loop**, which in each iteration:

- generates an event e from the current state s (initially s_0) and
- updates the environment state by invoking $\psi(s, e)$

The event generation is what *controls* the environment and is usually governed by entities (players) interacting with the environment and other random events (weather).

Environments

The Environment Loop - Example Rock Paper Scissors

Think of an environment where two players play the game Rock, Paper, Scissors.

The environment instance consists of

- 4 states (depending on the encoding): An initial state s_0 and 3 final states w , d , l to encode the result from first player's view,
- 9 events (each combination of players' moves), and
- a state transition function that encodes the game logic in s_0

This is one of the simplest environments possible (only one real event occurs).

Environments

The Environment Loop

The state transition function ψ is a *partial* function, because

- ① there can be **final states** in which no more events can occur and
- ② not all events may be processable in all states.

For convenience, we will assume that environments come with two more functions:

- ① APPLICABLE: $S \times E \rightarrow \{0, 1\}$ with $\text{APPLICABLE}(s, e) = 1$ iff $\psi(s, e)$ is defined.
- ② FINAL: $S \rightarrow \{0, 1\}$ with $\text{FINAL}(s) = 0$ iff $\exists e \in E : \text{APPLICABLE}(s, e) = 1$.

The environment loop can use these functions to stop or break upon invalid events.

Environments

Properties

An environment can be

- deterministic vs non-deterministic (property of ψ)
- static vs dynamic (property of the environment loop)
- discrete vs continuous (property of S and E)
- atomic (often called episodic) vs sequential (property of ψ)

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Agents

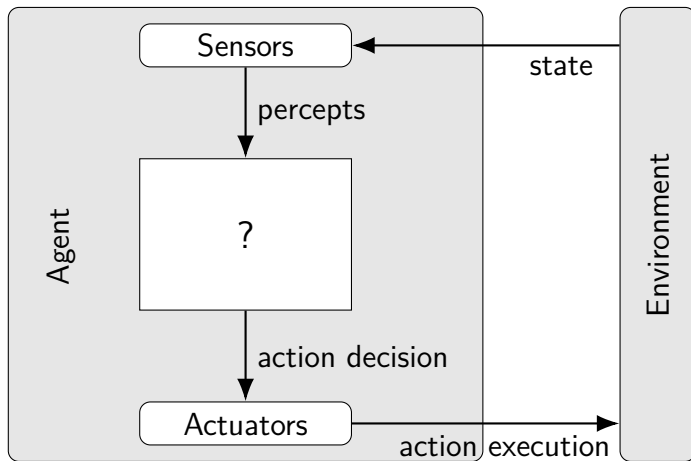
An **agent** is anything that can perceive (parts of) the state of an environment and influence events in it.

To do this, agents have

- **sensors** to capture **percepts** of a space P of its environment, and
- **actuators** to conduct **actions** of a space A in its environment.

Agents make **autonomous decisions** about which actions to perform based on the **history of percepts** (no intelligence without deliberation).

Agents and Environments



Agents

Connection Between Environments and Action

Mathematically speaking, percepts are a function of the environment state.

So whenever the environment advances into a new state s_t , the sensors generate a percept $p_t = p(s_t)$ at that point of time.

The agent then deliberates and picks an action $a_t \in A$.

Agents

Agent Function

From the outside, the agent's behavior can be described through an **agent function**:

$$f : P^* \rightarrow A$$

The agent function only says *what* the agent does but not how it decides its actions.

But this is a purely theoretical function: In practice,

- ① sensors will create a percept p_t of the current state s_t ,
- ② this percept will be passed to a decision making unit, and
- ③ the percept will be forgotten.

In many situations, the agent will hold no record of the exact percept sequence.

Agents

Agent Program

The factual realization of the agent function is called the **agent program**.

The agent program only takes the **current percept** as an input, because there is nothing outside the agent that is able to memorize the history.

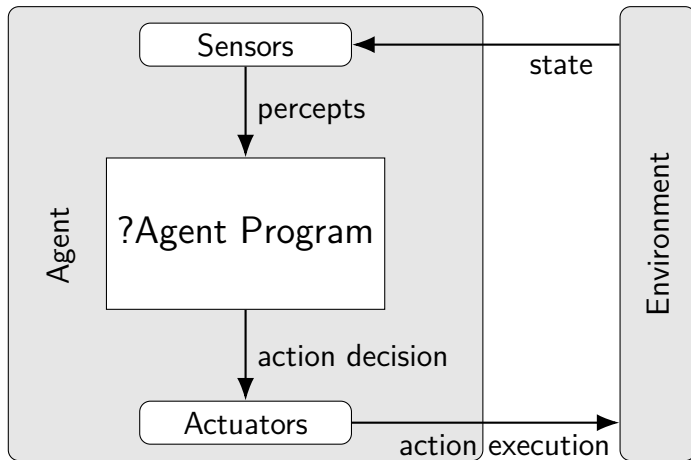
So the agent program is an *implementation* of a function

$$f : P \rightarrow A$$

and its outputs can be implemented through actuators in the environment.

Agents

The Agent Program



Areas of Intelligence in the Agent Program

According to the 3 components in the agent, there are 3 areas of intelligence:

- Percept Level: How I *interpret* raw signals received from sensors.
- Decision Level: How I *want* my interaction with the world to look like.
- Action Level: How I *implement* my interaction.



Agents

Properties of Agents

The agent setup of an environment also has several properties:

- Number of agents (one or many)
- For multiple agents: competitive/cooperative/teams
- Knowledge about environment physics (known vs unknown)
- Ability to observe the environment (fully vs partially)
- Ability to enforce actions (yes or no)

Agents

Examples for Properties

- Order a Pizza: non-deterministic, static, discrete, atomic/episodic, single-agent, environment unknown but observable
- Chess/Go: deterministic, static, discrete, sequential, multi-agent, environment known and observable
- Medical Treatment: non-deterministic, dynamic, continuous, atomic, single-agent, environment semi-known but only partially observable
- Autonomous Driving: non-deterministic, dynamic, continuous, sequential, multi-agent, environment known but only partially observable

Agents

Our Scope

The interaction between agents and environment can be very complex.

In this course, we will make the following simplifications:

- no sensing is necessary (this is covered in computer vision, sensor networks, etc.)
- no actuation is necessary (this is covered in robotics)

Agents

Our Scope

In this simplified scenario, the the environment loop can be implemented as follows:

- take current state s_t and use it as a percept
- compile action vector $\mathbf{a}_t := (a_t^1, a_t^2, \dots, a_t^k)$, where
 - ▶ k is the number of agents and
 - ▶ a_t^i is the i -th agent's picked action for state s_t
- use $\mathbf{a}_t$ as the event and update environment via $\psi(s_t, \mathbf{a}_t)$

So there are no effects that influence the environment other than the agents' actions.

If only one agent plays, use a dummy action for the others.

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Rational and Intelligent Agents

Up to now, we have not defined any objectives:

Agents can just move inside the environment forever without any purpose or goals.

In reality, some behaviors of agents are better (maybe more intelligent) than others.

How can we say whether an agent is intelligent?

Rational and Intelligent Agents

Rational Agents

A first requirement for an agent to be intelligent is that the chosen actions make sense, i.e., choices are **rational**.

What is rational depends on

- a **performance measure** that defines the criterion of success
- the actions the agent can perform
- the agent's knowledge of the environment and other agents' intended actions when deciding

An agent is then rational if it selects an action that maximizes the expected performance given its knowledge and the available actions.

Rational and Intelligent Agents

Deep Blue vs. Kasparov (1997) - Rational Behavior by Large Databases and Comparison



Rational and Intelligent Agents

Main Building Blocks for an Agent Program that Achieves Rational Behavior

The main ingredients to build the agent program upon are:

- **knowledge**
- **inference** (deriving “new” knowledge from existing knowledge)
- **learning** (updates of the knowledge based on percepts)

Learning is needed to make agents independent of hard-wired knowledge.

Typical procedure: The agent program adopts learning to incorporate the percepts into the knowledge base. The new action is then directly dictated by the knowledge or is inferred from it via inference.

To a certain degree, these building blocks are substitutable by each other.

Rational and Intelligent Agents

Intelligent Agents

Until two decades ago, rationality was all we cared about.

Many agents were not called intelligent even if they acted rationally, because ultimately all behavior was pre-determined by the developers.

True intelligence (hard to grasp) requires some kind of autonomous learning.

An **intelligent agent** is one in which learning has enabled significant *generalized knowledge* not explicitly encoded that can be used for inference of rational decisions.

Rational and Intelligent Agents

Lee Sedol vs. AlphaGo (2016) - Intelligent Agent by learned behavior

